Week 02 teacher guide

Ratios and matching amounts

PowerPoint slides 107-201 | Student workbook pp. 14-24

A ratio compares two amounts in a chosen order. Equivalent ratios scale both amounts together. Tables and double number lines keep matching amounts connected.

The seven days

Day 008 · pp. 14-16 · Saturday · 20 questions,including 2 challenges
Equivalent ratios

Day 009 · pp. 17-19 · Sunday · 20 questions,including 2 challenges
Ratio tables and comparisons

Day 010 · p. 20 · Monday · 20 questions,including 2 challenges
Ratios and matching amounts

Day 011 · p. 21 · Tuesday · 20 questions,including 2 challenges
Ratios and matching amounts

Day 012 · p. 22 · Wednesday · 25 questions,including 2 challenges
Ratios and matching amounts

Day 013 · p. 23 · Thursday · 20 questions,including 2 challenges
Ratios and matching amounts

Day 014 · p. 24 · Friday · 20 questions,including 2 challenges
Weekly check

Use one day at a time

Each weekend lesson uses a three-page packet; each weekday uses one question page. Work on separate plain paper, labeling the Day number and item IDs. Weekend lessons and ordinary weekdays contain 20 questions including C1 and C2; Wednesday contains 25. Allow 30-45 minutes on ordinary weekdays and 40-55 minutes on Wednesday. Split a demanding day into two sittings if needed.

The weekend packet includes models and worked examples, or a scheduled assessment alternative. Read a worked example after an attempt. Keep answer-review slides out of view until the corresponding work is complete. Weekly answer review is optional after Friday and does not add to the 80-minute weekend lesson.

Built-in review

Day 010: Ratios and rates (9, 10, 11, 12)

Day 011: Ratios and rates (9, 10, 11, 12)

Day 012: Weekly puzzle (13)

Day 013: Fractions: compare and combine (9, 10, 11, 12)

On Friday, record routine accuracy out of 18 separately from the two challenge explanations. For each challenge, note whether the learner connected the steps, used a valid model or calculation, and justified the conclusion. Revisit a missed model, then check with a fresh item later.

Saturday. Equivalent ratios

Day 008 · pp. 14-16 | Slides 107-136

Goal: I can keep a ratio unchanged while making more or less of a mixture.

Check first: Multiplication, division, and equal groups.

Materials: pencil and plain paper; use the projected models. Sketch strips, grids, or nets as needed; make any manipulatives from plain paper.

80 minutes: recall 7; launch and explore 18; model and guided practice 20; compare methods 10; challenges 20; summary and exit 5. Allow up to 10 extra minutes for discussion. Assessment alternatives replace routine practice.

Follow the numbered slide order below. The timing budgets span the named phases even when guided practice or discussion returns later. On busy weekends, split the lesson into two sittings and resume at the next slide.

Teaching sequence

Slides 107-108 · Orient

Read the goal. Ask what is familiar; name the quantity or relationship in focus.

Slides 109-110 · Recall · R1, R2, R3, R4

Try from memory, then check. Use one error for a quick reminder; keep this brief.

Slides 111-112 · Explore · P1, P2, P3

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 113 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 114-115 · Explore · P4, P5, P6

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 116-118 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 119 · Guided practice · F

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 120 · Compare methods

Connect diagram and calculation. Ask where each quantity appears and why both methods preserve it.

Slides 121 · Discuss an error · D

Locate the first incorrect step. Explain why it is tempting; repair it with a model or counterexample.

Slides 122 · Language

Attach each term to a model. Ask for an example and a non-example.

Slides 123-126 · Guided practice · X1, X2, X3, X4

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 127-128 · Deepening

Use the investigation below. Predict, calculate, and explain the structural reason. Budget about 8 minutes.

Slides 129-132 · Challenges · C1, C2

Budget about 10 minutes per challenge. Require linked reasoning; give a single hint after an independent attempt.

Slides 133 · Summary · S

Ask for one sentence or annotated example in the learner’s own words. Listen for the main relationship.

Slides 134 · Exit · E

Keep answers hidden. Collect an independent response and its reasoning before helping.

Slides 135 · Weekday bridge

Point to the next five days: 18 routines plus 2 challenges, or 23 routines plus 2 challenges on Wednesday.

Slides 136 · Tutor check

After the exit, use the hidden tutor slide to identify secure understanding, a small repair, or needed re-teaching.

Saturday. Discussion and support

Investigation T: Two equivalent-ratio methods meet

Watch and discuss; this is a projected investigation, not an extra assigned question.

A mix uses 3 cups oats for 4 cups seeds. Find seeds for $7\frac{1}{2}$ cups oats in two ways: scale by one number, and combine full and partial batches. Explain why the methods agree.

1. Write the known ratio 3 : 4.
2. Scale both entries by $\frac{5}{2}$.
3. Build the same amounts as two batches plus half a batch.
4. Connect the sum of the batches to the scale factor.

Ask: What does the half batch mean in both columns? Why must both quantities change together?
If stuck: First make 6 cups oats, then identify what fraction of one batch fills the gap to $7\frac{1}{2}$.

Expected: 10 cups seeds. The scale factor is $\frac{5}{2}$; two full batches plus one half batch gives the same amount.

C1 · Challenge discussion

Ask: Why must the same number of groups apply to both colors?
If stuck: Circle groups of 3 red tiles, then match each group to 5 blue.
Expected: 30 blue tiles needed; 2 remain

C2 · Challenge discussion

Ask: How could you build $2\frac{1}{2}$ batches without using a decimal rule?

If stuck: Make two full batches and one half batch.
Expected: 3 cups seeds remain

What to notice

Says four red for every three blue. → Ask for the reversed ratio and compare meanings.

Repeats or partitions a complete group. → Ask which factor acts on both amounts.

Common error: Adds the same count to both colors. Repair: Build identical batches.

Sunday. Ratio tables and comparisons

Day 009 · pp. 17-19 | Slides 137-165

Goal: I can use tables, double number lines, and equal totals to solve ratio problems.

Check first: Scaling both amounts in a ratio.

Materials: pencil and plain paper; use the projected models. Sketch strips, grids, or nets as needed; make any manipulatives from plain paper.

80 minutes: recall 7; launch and explore 18; model and guided practice 20; compare methods 10; challenges 20; summary and exit 5. Allow up to 10 extra minutes for discussion. Assessment alternatives replace routine practice.

Follow the numbered slide order below. The timing budgets span the named phases even when guided practice or discussion returns later. On busy weekends, split the lesson into two sittings and resume at the next slide.

Teaching sequence

Slides 137-138 · Orient

Read the goal. Ask what is familiar; name the quantity or relationship in focus.

Slides 139-140 · Recall · R1, R2, R3, R4

Try from memory, then check. Use one error for a quick reminder; keep this brief.

Slides 141 · Explore · P1

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 142 · Compare methods · P2

Connect diagram and calculation. Ask where each quantity appears and why both methods preserve it.

Slides 143 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 144-145 · Explore · P3, P4, P5

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 146-148 · Model

Reveal one step at a time after the attempt. Ask what each number or mark represents before moving on.

Slides 149 · Guided practice · F

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 150 · Compare methods

Connect diagram and calculation. Ask where each quantity appears and why both methods preserve it.

Slides 151 · Discuss an error · D

Locate the first incorrect step. Explain why it is tempting; repair it with a model or counterexample.

Slides 152 · Language

Attach each term to a model. Ask for an example and a non-example.

Slides 153 · Explore · M1, M2, M3

Let the learner start first. Ask for a drawing or first step, then compare quantities, units, and strategies.

Slides 154-155 · Guided practice · X1, X2

Let the learner finish and justify each missing step. Reduce hints after one successful explanation.

Slides 156-157 · Deepening

Use the investigation below. Predict, calculate, and explain the structural reason. Budget about 8 minutes.

Slides 158-161 · Challenges · C1, C2

Budget about 10 minutes per challenge. Require linked reasoning; give a single hint after an independent attempt.

Slides 162 · Summary · S

Ask for one sentence or annotated example in the learner's own words. Listen for the main relationship.

Slides 163 · Exit · E

Keep answers hidden. Collect an independent response and its reasoning before helping.

Slides 164 · Weekday bridge

Point to the next five days: 18 routines plus 2 challenges, or 23 routines plus 2 challenges on Wednesday.

Slides 165 · Tutor check

After the exit, use the hidden tutor slide to identify secure understanding, a small repair, or needed re-teaching.

Sunday. Discussion and support

Investigation T: Same water or same total?

Watch and discuss; this is a projected investigation, not an extra assigned question.

Mix A uses 3 cups concentrate for 4 cups water. Mix B uses 4 for 7. Compare strength using the same amount of water, then using the same total amount of mixture.

1. Name what each ratio compares.
2. Make the water amount 28 in both recipes.
3. Make the total mixture 77 in both recipes.
4. Explain why both comparisons give the same conclusion.

Ask: Why can the comparison amount be water or total mixture, but cannot silently switch between them?

If stuck: Label every denominator before choosing a common multiple.

Expected: Mix A is stronger. At 28 cups water, A has 21 cups concentrate and B has 16. At 77 cups total, A has 33 cups concentrate and B has 28.

C1 · Challenge discussion

Ask: Which quantity changes, and which stays fixed?

If stuck: Draw seven equal boxes before adding the extra gold beads.

Expected: Originally 18 gold and 24 silver; new ratio 24 : 24 = 1 : 1

C2 · Challenge discussion

Ask: Why does the same amount of water make this comparison fair?

If stuck: Use 15 as the common water amount.

Expected: Mix A: 10 cups concentrate; Mix B: 9 cups; Mix A is stronger

What to notice

Keeps matching values in the same row. → Ask what one complete row means.

Finds a common water amount or compares part fractions. → Accept either method and require a shared comparison unit.

Common error: Compares mixtures by subtraction. Repair: Use the same amount of water in both mixtures.

Slide-order index

Follow this numerical order. The session plans explain how to teach each phase; full scripts remain in PowerPoint speaker notes. Item IDs match the student workbook.

Saturday

107. Week 2, Saturday
Watch and discuss

108. Today
Watch and discuss

109. Recall
Day 008 · Workbook p. 14 · R1, R2, R3, R4

110. Recall: answers
Day 008 · Workbook p. 14 · R1, R2, R3, R4

111. Read a picture and preserve order
Day 008 · Workbook p. 14 · P1

112. Scale up by a whole number
Day 008 · Workbook p. 14 · P2, P3

113. The model
Watch and discuss

114. Non-whole scale factor
Day 008 · Workbook pp. 14, 15 · P4, P5

115. Missing value in groups
Day 008 · Workbook p. 15 · P6

116. Worked example · 1 of 3
Watch and discuss · worked example

117. Worked example · 2 of 3
Watch and discuss · worked example

118. Worked example · 3 of 3
Watch and discuss · worked example

119. Your turn
Day 008 · Workbook p. 15 · F

120. Two ways to see it
Watch and discuss · compare the same example

121. A claim to check
Day 008 · Workbook p. 15 · D

122. Words and meaning
Watch and discuss

123. Guided practice · X1
Day 008 · Workbook p. 15 · X1

124. Guided practice · X2
Day 008 · Workbook p. 15 · X2

125. Guided practice · X3
Day 008 · Workbook p. 15 · X3

126. Guided practice · X4
Day 008 · Workbook p. 15 · X4

127. Two equivalent-ratio methods meet
Watch and discuss · draw on separate paper

128. Two equivalent-ratio methods meet
Watch and discuss · draw on separate paper

129. Challenge C1
Day 008 · Workbook p. 16 · C1

130. Challenge C1 · Discuss
Day 008 · Workbook p. 16 · C1

131. Challenge C2
Day 008 · Workbook p. 16 · C2

132. Challenge C2 · Discuss
Day 008 · Workbook p. 16 · C2

133. In your own words
Day 008 · Workbook p. 16 · S

134. Exit ticket
Day 008 · Workbook p. 16 · E

135. The week ahead
Days 010-014 · Workbook pp. 20-24

136. Tutor reference
Tutor only;hidden in slide show

Sunday

137. Week 2, Sunday
Watch and discuss

138. Today
Watch and discuss

139. Recall
Day 009 · Workbook p. 17 · R1, R2, R3, R4

140. Recall: answers
Day 009 · Workbook p. 17 · R1, R2, R3, R4

141. One missing table value
Day 009 · Workbook p. 17 · P1

142. Solve the same problem two ways
Day 009 · Workbook p. 17 · P2

143. The model
Watch and discuss

144. Compare using a common amount
Day 009 · Workbook p. 17 · P3, P4

145. Ratio points in the first quadrant
Day 009 · Workbook p. 18 · P5

146. Worked example · 1 of 3
Watch and discuss · worked example

147. Worked example · 2 of 3
Watch and discuss · worked example

148. Worked example · 3 of 3
Watch and discuss · worked example

149. Your turn
Day 009 · Workbook p. 18 · F

150. Two ways to see it
Watch and discuss · compare the same example

151. A claim to check
Day 009 · Workbook p. 18 · D

152. Words and meaning
Watch and discuss

153. Mixed review
Day 009 · Workbook p. 18 · M1, M2, M3

154. Guided practice · X1
Day 009 · Workbook p. 18 · X1

155. Guided practice · X2
Day 009 · Workbook p. 18 · X2

156. Same water or same total?
Watch and discuss · draw on separate paper

157. Same water or same total?
Watch and discuss · draw on separate paper

158. Challenge C1
Day 009 · Workbook p. 19 · C1

159. Challenge C1 · Discuss
Day 009 · Workbook p. 19 · C1

160. Challenge C2
Day 009 · Workbook p. 19 · C2

161. Challenge C2 · Discuss
Day 009 · Workbook p. 19 · C2

162. In your own words
Day 009 · Workbook p. 19 · S

163. Exit ticket
Day 009 · Workbook p. 19 · E

164. The week ahead
Days 010-014 · Workbook pp. 20-24

165. Tutor reference
Tutor only;hidden in slide show

Answer review

166. Week 2 answers
Next lesson: slide 202. Use this review section after Friday.

167. Equivalent ratios · 1 of 5
Day 008 · Workbook p. 14 · Items R1, R2, R3, R4

168. Equivalent ratios · 2 of 5
Day 008 · Workbook pp. 14, 15 · Items P1, P2, P3, P4, P5

169. Equivalent ratios · 3 of 5
Day 008 · Workbook p. 15 · Items P6, F, D, X1, X2

170. Equivalent ratios · 4 of 5
Day 008 · Workbook pp. 15, 16 · Items X3, X4, S, E

171. Equivalent ratios · 5 of 5
Day 008 · Workbook p. 16 · Items C1, C2

172. Ratio tables and comparisons · 1 of 5
Day 009 · Workbook p. 17 · Items R1, R2, R3, R4

173. Ratio tables and comparisons · 2 of 5
Day 009 · Workbook pp. 17, 18 · Items P1, P2, P3, P4, P5

174. Ratio tables and comparisons · 3 of 5
Day 009 · Workbook p. 18 · Items F, D, M1, M2

175. Ratio tables and comparisons · 4 of 5
Day 009 · Workbook pp. 18, 19 · Items M3, X1, X2, S

176. Ratio tables and comparisons · 5 of 5
Day 009 · Workbook p. 19 · Items E, C1, C2

177. Ratios and matching amounts · 1 of 5
Day 010 · Workbook p. 20 · Items 1, 2, 3, 4, 5

178. Ratios and matching amounts · 2 of 5
Day 010 · Workbook p. 20 · Items 6, 7, 8, 9, 10

179. Ratios and matching amounts · 3 of 5
Day 010 · Workbook p. 20 · Items 11, 12, 13, 14, 15

180. Ratios and matching amounts · 4 of 5
Day 010 · Workbook p. 20 · Items 16, 17, 18, C1

181. Ratios and matching amounts · 5 of 5
Day 010 · Workbook p. 20 · Items C2

182. Ratios and matching amounts · 1 of 5
Day 011 · Workbook p. 21 · Items 1, 2, 3, 4, 5

183. Ratios and matching amounts · 2 of 5
Day 011 · Workbook p. 21 · Items 6, 7, 8, 9, 10

184. Ratios and matching amounts · 3 of 5
Day 011 · Workbook p. 21 · Items 11, 12, 13, 14

185. Ratios and matching amounts · 4 of 5
Day 011 · Workbook p. 21 · Items 15, 16, 17, 18, C1

186. Ratios and matching amounts · 5 of 5
Day 011 · Workbook p. 21 · Items C2

187. Ratios and matching amounts · 1 of 5
Day 012 · Workbook p. 22 · Items 1, 2, 3, 4, 5

188. Ratios and matching amounts · 2 of 5
Day 012 · Workbook p. 22 · Items 6, 7, 8, 9, 10

189. Ratios and matching amounts · 3 of 5
Day 012 · Workbook p. 22 · Items 11, 12, 13, 14, 15

190. Ratios and matching amounts · 4 of 5
Day 012 · Workbook p. 22 · Items 16, 17, 18, 19, 20

191. Ratios and matching amounts · 5 of 5
Day 012 · Workbook p. 22 · Items 21, 22, 23, C1, C2

192. Ratios and matching amounts · 1 of 5
Day 013 · Workbook p. 23 · Items 1, 2, 3, 4, 5

193. Ratios and matching amounts · 2 of 5
Day 013 · Workbook p. 23 · Items 6, 7, 8, 9

194. Ratios and matching amounts · 3 of 5
Day 013 · Workbook p. 23 · Items 10, 11, 12, 13

195. Ratios and matching amounts · 4 of 5
Day 013 · Workbook p. 23 · Items 14, 15, 16, 17, 18

196. Ratios and matching amounts · 5 of 5
Day 013 · Workbook p. 23 · Items C1, C2

197. Weekly check · 1 of 5
Day 014 · Workbook p. 24 · Items 1, 2, 3, 4, 5

198. Weekly check · 2 of 5
Day 014 · Workbook p. 24 · Items 6, 7, 8, 9

199. Weekly check · 3 of 5
Day 014 · Workbook p. 24 · Items 10, 11, 12, 13, 14

200. Weekly check · 4 of 5
Day 014 · Workbook p. 24 · Items 15, 16, 17, 18, C1

201. Weekly check · 5 of 5
Day 014 · Workbook p. 24 · Items C2

Complete weekly answers

Match Day,item ID,and page before checking. Equivalent correct forms and valid alternative methods are welcome. Computational keys are compact; challenges and conceptual items include reasoning. No challenge is complete with an unsupported answer alone.

Day 008 · Equivalent ratios

Workbook pp. 14-16 · assigned set

R1 · F-DIV
$\frac{1}{6}$ liter

R2 · F-COMP
$\frac{5}{8} > \frac{3}{5}$

Reason: Rename as twenty-five fortieths and twenty-four fortieths.

R3 · F-COMP
$\frac{1}{3} > \frac{1}{4}$

Reason: With the same numerator, the fraction with larger pieces is larger.

R4 · F-MULT
$\frac{3}{5}$

P1 · R-RATIO
4 : 3; 3 : 4

P2 · R-EQUIV
9 nuts

P3 · R-EQUIV
4 : 3

Reason: Make five equal groups. Each has four red and three blue.

P4 · R-EQUIV
6 cups

P5 · R-EQUIV
6 cups yellow

P6 · R-EQUIV
12 builders

F · R-EQUIV
2 batches; 4 cups seeds

Reason: Check each step against the projected worked example; require the learner to explain the operation or relationship.

D · R-EQUIV
No. Two copies would give 8 : 6, not 7 : 6.

Reason: She uses equal additions rather than equal multiplication. Build two full copies of the original group and compare.

X1 · R-EQUIV
14

X2 · R-EQUIV
3 : 7

X3 · R-EQUIV
15 cups

X4 · R-EQUIV
No; cross-products 30 and 40 differ.

Reason: Equivalent ratios use one multiplicative scale factor. Adding 5 to both amounts does not preserve it.

S · EXPLAIN
Answers vary. Equivalent ratios multiply or divide both quantities by the same factor.

Reason: Accept an accurate explanation supported by a matching example.

E · R-EQUIV
21 green beads

Reason: Twelve white beads are three groups of four. Use three groups of seven green.

C1 · R-EQUIV
30 blue tiles needed; 2 remain

Reason: 18 red tiles make 6 groups of 3. Each group needs 5 blue, so $6 \times 5 = 30$ blue tiles. Subtract 30 from 32 to find 2 unused tiles.

C2 · R-EQUIV
3 cups seeds remain

Reason: The oats scale from 4 to 10 by $\frac{5}{2}$. Seeds must scale by $\frac{5}{2}$ too: $6 \times \frac{5}{2} = 15$ cups. $18 - 15 = 3$ cups remain.

Day 009 · Ratio tables and comparisons

Workbook pp. 17-19 · assigned set

R1 · F-LINE
At 1 $\frac{3}{4}$, the seventh quarter-step from 0

R2 · F-SUB
$\frac{5}{12}$

R3 · F-SCALE

Smaller; 6

Reason: Two thirds takes less than all nine.

R4 · F-MULT

2

P1 · R-TABLE

10 cups water

P2 · R-TABLE

25 cups water

P3 · R-COMP

Mix A

Reason: Double A to get 6 syrup : 8 water. Compare 6 syrup with 5 syrup.

P4 · R-WHOLE

14 cups yellow and 21 cups blue

P5 · R-GRAPH

Yes

Reason: Each y-value is three times its x-value. Zero bags have zero marbles.

F · R-COMP

6 per part; 24 green; 6 white

Reason: Check each step against the projected worked example; require the learner to explain the operation or relationship.

D · R-COMP

There are five parts in all. Yellow is $\frac{2}{5}$ of the paint.

Reason: He uses the other part as the whole. Join the two yellow parts and three blue parts into one total strip.

M1 · F-SCALE

Larger; 8

Reason: Four thirds takes one whole six plus one third of six.

M2 · F-DIV

6 pieces

M3 · F-COMP

$\frac{7}{8}$

Reason: The gaps from 1 are one eighth and one fifth. One eighth is smaller.

X1 · R-PART1

15 gold; 25 silver

X2 · R-PART1

$\frac{5}{8}$

S · EXPLAIN

Answers vary. A ratio table records equal groups. Comparing mixtures needs a common amount or a common total.

Reason: Accept an accurate explanation supported by a matching example.

E · R-COMP

Mix A

Reason: A is $\frac{4}{7}$ juice; B is $\frac{6}{11}$. Compare $\frac{44}{77}$ and $\frac{42}{77}$.

C1 · R-PART1

Originally 18 gold and 24 silver; new ratio 24 : 24 = 1 : 1

Reason: Seven equal parts share 42 beads, so each part has 6. Gold = 18 and silver = 24. Adding 6 gold gives 24 of each color; adding only one color changes the ratio.

C2 · R-COMP

Mix A: 10 cups concentrate; Mix B: 9 cups; Mix A is stronger

Reason: Multiply Mix A by 5 and Mix B by 3 to get the same 15 cups water. More concentrate for that fixed amount of water means a stronger mixture.

Day 010 · Ratios and matching amounts

Workbook p. 20 · assigned set

1 · N-FACT

12

2 · N-FACT

28

3 · N-FACT

16

4 · N-FACT

36

5 · F-COMP

$\frac{1}{3} > \frac{1}{4}$

Reason: With the same numerator, the fraction with larger pieces is larger.

6 · D-OPS

2.75

7 · N-FACT

25

8 · R-EQUIV

3 cups

9 · R-RATIO · review

5 : 2; 2 : 5

Reason: Keep the requested first color first.

10 · R-EQUIV · review

Yes

Reason: Multiply both numbers in 6 : 10 by one and one half.

11 · R-EQUIV · review

No

12 · R-EQUIV · review

24 runners

13 · R-EQUIV

No. Two copies would give 8 : 6, not 7 : 6.

14 · R-EQUIV

6

15 · R-EQUIV

3 : 4

16 · R-EQUIV

15 cups

17 · R-PART1

30 gold; 36 silver

18 · R-PART1

$\frac{6}{13}$

C1 · R-EQUIV

7 patterns; 0 red and 3 green flags unused

Reason: Red allows 28 ÷ 4 = 7 patterns. Seven patterns need 49 green flags, leaving 3 green. An eighth pattern would require 32 red, so it cannot be made.

C2 · R-EQUIV

$\frac{9}{2}$ cups blue; $\frac{15}{2}$ cups total; scale factor $\frac{3}{4}$

Reason: Three is $\frac{3}{4}$ of 4. Multiply 6 by $\frac{3}{4}$ to get $\frac{9}{2}$ cups blue. Add 3 cups yellow to get $\frac{15}{2}$ cups total.

Day 011 · Ratios and matching amounts

Workbook p. 21 · assigned set

1 · F-EQUIV

0.5; 50%

2 · F-EQUIV

0.25; 25%

3 · F-EQUIV

0.75; 75%

4 · F-EQUIV

0.125; 12.5%

5 · F-COMP

$\frac{1}{3} > \frac{1}{4}$

Reason: With the same numerator, the fraction with larger pieces is larger.

6 · N-SCALE

3

7 · N-FACT

12

8 · R-WHOLE

8 yellow; 20 purple

9 · R-TABLE · review

21

10 · R-TABLE · review

32 laps

11 · R-GRAPH · review

Yes

Reason: Every vertical value is twice the horizontal value.

12 · R-WHOLE · review

$\frac{3}{8}$

13 · R-COMP

There are five parts in all. Yellow is $\frac{2}{5}$ of the paint.

14 · R-EQUIV

16

15 · R-EQUIV

5 : 4

16 · R-EQUIV

21 cups

17 · R-PART1

12 gold; 9 silver

18 · R-PART1

$\frac{3}{11}$

C1 · R-TABLE

(8, 54) should be (8, 56); zero bags gives (0, 0)

Reason: The first two pairs each have 7 marbles per bag, since 14 ÷ 2 = 35 ÷ 5 = 7. Eight bags must have 56 marbles. Zero equal groups contain zero marbles.

C2 · R-PART1
Team A 16 red; Team B 15 red. $\frac{2}{5} > \frac{3}{8}$

Reason: Team A has 5 parts, 8 ribbons each, so 16 red. Team B has 8 parts, 5 ribbons each, so 15 red. The same total makes the one-ribbon difference a fair comparison.

Day 012 · Ratios and matching amounts

Workbook p. 22 · assigned set

1 · N-SCALE
42

2 · N-SCALE
4

3 · N-SCALE
9

4 · N-SCALE
12

5 · R-EQUIV
Yes
Reason: Multiply both numbers in 6 : 10 by one and one half.

6 · R-EQUIV
No

7 · F-MULT
2

8 · F-ADD
$\frac{5}{6}$

9 · D-OPS
4.75

10 · G-VOL
36 cm³

11 · N-FACT
14

12 · G-VOL
42 cm³

13 · PUZZLE · review
15 red and 10 blue

14 · R-EQUIV
30

15 · R-EQUIV
2 : 4

16 · R-EQUIV
12 cups

17 · R-PART1
15 gold; 25 silver

18 · R-PART1
$\frac{5}{8}$

19 · R-TABLE
35; 30

20 · R-EQUIV
No; cross-products 12 and 16 differ.
Reason: Equivalent ratios use one multiplicative scale factor. Adding 2 to both amounts does not preserve it.

21 · R-COMP
Mix B
Reason: Double Mix A to get 6 cups fruit for 10 cups yogurt. Mix B has one more cup fruit at that same yogurt amount.

22 · R-GRAPH
(0, 0), (4, 16), (5, 20); yes

23 · F-MULT
4 kilograms

C1 · R-PART1
4 blue beads must become red
Reason: There are 9 parts of 8 beads: 32 red and 40 blue. Equal counts would be 36 and 36. Recoloring 4 blue as red increases red by 4 and decreases blue by 4.

C2 · R-EQUIV
n = 8; (14, 18) is not equivalent; the correct pair is (14, 21)
Reason: Since 12 is 4 times 3, n = 4 × 2 = 8. Adding 6 gives 14 : 18 = 7 : 9, not 2 : 3. To make the first entry 14, multiply 2 : 3 by 7 to get 14 : 21.

Day 013 · Ratios and matching amounts

Workbook p. 23 · assigned set

1 · N-EST
400

2 · N-EST
5

3 · N-EST
About 100

4 · N-EST
About 20

5 · R-EQUIV

Yes

Reason: Multiply both numbers in 6 : 10 by one and one half.

6 · R-EQUIV

No

7 · R-EQUIV

4 : 7

8 · F-DIV

$\frac{1}{18}$; stories vary

9 · F-ADD · review

$\frac{5}{6}$

10 · F-ADD · review

$\frac{11}{12}$

11 · F-ADD · review

$\frac{7}{12}$

12 · F-ADD · review

$\frac{3}{4}$

13 · R-EQUIV

24 runners

14 · R-COMP

There are five parts in all. Yellow is $\frac{2}{5}$ of the paint.

15 · R-EQUIV

18

16 · R-EQUIV

4 : 4

17 · R-EQUIV

18 cups

18 · R-PART1

6 gold; 9 silver

C1 · R-EQUIV

8 cups sugar used; 2 left; 28 cups ingredients

Reason: Twenty cups flour is four 5-cup groups. Match with four 2-cup groups of sugar: 8 cups. Ten minus 8 leaves 2; 20 + 8 = 28 cups in the batch.

C2 · R-PART1

16 triangles and 21 squares; 16 : 21

Reason: Seven full patterns use 35 shapes: 14 triangles and 21 squares. The next two shapes are triangles, giving 16 and 21. A partial pattern changes the overall ratio.

Day 014 · Weekly check

Workbook p. 24 · assigned set

1 · R-EQUIV

Yes

Reason: Multiply both numbers in 6 : 10 by one and one half.

2 · R-EQUIV

No

3 · R-EQUIV

4 : 7

4 · R-EQUIV

4 : 7

5 · F-COMP

$\frac{2}{4} > \frac{2}{5}$

Reason: With the same numerator, the fraction with larger pieces is larger.

6 · F-LINE

At $1\frac{3}{4}$, the seventh quarter-step from 0

7 · F-MULT

$\frac{3}{5}$

8 · D-OPS

6.75

9 · G-VOL

90 cm³

10 · R-EQUIV

No. Two copies would give 8 : 6, not 7 : 6.

11 · R-EQUIV

42

12 · R-EQUIV

2 : 8

13 · R-EQUIV

12 cups

14 · R-PART1

25 gold; 25 silver

15 · R-PART1

$\frac{1}{2}$

16 · R-TABLE

21; 12

17 · R-EQUIV

No; cross-products 32 and 36 differ.

Reason: Equivalent ratios use one multiplicative scale factor. Adding 2 to both amounts does not preserve it.

18 · R-COMP

Mix B

Reason: Double Mix A to get 6 cups fruit for 10 cups yogurt. Mix B has one more cup fruit at that same yogurt amount.

C1 · R-COMP

A needs 15 cups lemon, B needs 14; A needs 1 cup more

Reason: Scale A by 5 and B by 7 to get 35 cups water. Lemon amounts become 15 and 14. The comparison uses the same water amount.

C2 · R-PART1

10 adults and 25 children; new ratio 10 : 35 = 2 : 7

Reason: Seven equal parts split 35 people into groups of 5. There are 10 adults and 25 children. Ten more children make 35 children; simplify 10 : 35 by dividing both terms by 5.